# Served stats

Throughput, latency and quality of each served recipe, by model and hardware

2026-10-20 09:33 UTC

Rendered 2026-10-20 09:33 UTC from stats/*.json, the ledger every harness run and harness export adds to (python3 -m harness stats renders it again). Aggregate tok/s, greedy, 256-token replies, the recipe's own ruler; engine rounds from the server's own counters.

## Overview: by model and hardware

Aggregate tok/s as prose/code/structured/list; first token in seconds (one stream / the most streams).

| model | hardware | one stream | most streams: total | first token (s) | suite | measured |
|---|---|---|---|---|---|---|
| DeepSeek-V4.1-Flash-1017 EXL3 2.9bpw | 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE | 55/100/62/71 | 4: 106/162/112/179 | 0.19 / 0.73-0.99 | | 2026-10-18 |
| DeepSeek-V4.1-Flash-1017 EXL3 2.9bpw | 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE | 55/104/63/71 | 8: 135/203/144/203 | 0.19 / 1.35-1.84 | | 2026-10-18 |
| DeepSeek-V4.1-Flash-1017 EXL3 2.9bpw | 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE | 53/98/61/69 | 4: 106/158/111/176 | 0.20 / 0.75-1.00 | | 2026-10-18 |
| DeepSeek-V4.1-Flash-1017 EXL3 2.9bpw | 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE | 55/101/63/71 | 4: 104/156/109/173 | 0.19 / 0.41-0.64 | | 2026-10-17 |
| DeepSeek-V4.1-Flash-1017, 2/3-bit experts | 1x Apple M5 Ultra (256 GB unified), macOS | 70/88/80/94 | 4: 107/120/116/124 | 0.10 / 0.44-0.58 | tools 58/60, json 30/30, gsm8k 247/250, ifeval 129/150 | 2026-10-20 |
| GLM-5.3-Flash EXL3 4bpw | 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE | 42/60/57/58 | | 0.17 | json 30/30, ifeval 103/150 | 2026-10-17 |
| GLM-5.3-Flash EXL3 4bpw | 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE | 27/48/39/33 | 8: 78/115/98/110 | 0.18 / 1.49-1.83 | | 2026-10-17 |
| GLM-5.3-Flash EXL3 4bpw | 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE | -/-/-/- | 16: 107/120/119/130 | / 2.69-3.38 | | 2026-10-18 |
| GLM-5.3-Flash EXL3 4bpw | 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE | -/-/-/- | 16: 106/120/131/138 | / 1.32-1.62 | | 2026-10-18 |
| GLM-5.3-Flash EXL3 4bpw | 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE | -/-/-/- | 16: 117/134/148/159 | / 1.34-1.66 | | 2026-10-18 |
| GLM-5.3-Flash EXL3 4bpw | 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE | | | | json 30/30, ifeval 103/150 | 2026-10-17 |

| model | hardware | one stream | most streams: total | first token (s) | suite | measured |
|---|---|---|---|---|---|---|
| GLM-5.3-Flash EXL3 4bpw | 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE | 28/38/33/31 | 16: 124/156/150/145 | 0.18 / 8.50-17.20 | | 2026-10-16 |
| GLM-5.3-Flash EXL3 4bpw | 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE | 28/39/34/31 | 16: 86/101/100/102 | 0.18 / 3.23-3.72 | | 2026-10-17 |
| GLM-5.3-Flash, EXL3 4 bpw | 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE | 46/69/57/59 | | 0.18 | tools 59/60, json 30/30, gsm8k 246/250, humaneval 159/164, ifeval 118/150 | 2026-10-14 |
| GLM-5.3-Flash, MLX 4-bit | 1x Apple M5 Ultra (256 GB unified), macOS | 74/78/78/86 | | 0.11 | tools 57/60, json 30/30, gsm8k 246/250 | 2026-10-16 |
| Gemma 4 12B, MLX 4-bit + MTP assistant | 1x Apple M5 Ultra (256 GB unified), macOS | 140/223/205/248 | | 0.04 | tools 58/60, json 30/30, gsm8k 239/250 | 2026-10-16 |
| Gemma 4 12B, oQ8e (8-bit) + MTP assistant | 1x Apple M5 Ultra (256 GB unified), macOS | 93/142/146/158 | 4: 262/324/328/340 | 0.05 / 0.12-0.12 | | 2026-10-16 |
| Gemma 4 31B, MLX 4-bit + DFlash | 1x Apple M5 Ultra (256 GB unified), macOS | 87/158/157/130 | | 0.08 | | 2026-10-16 |
| Gemma 4 31B, oQ8e (8-bit) + DFlash | 1x Apple M5 Ultra (256 GB unified), macOS | 40/63/52/55 | 4: 110/173/142/180 | 0.10 / 0.24-0.27 | | 2026-10-16 |
| NVIDIA Nemotron 3.5 Bolt 74B-A9B (24 GB: IQ2_XXS experts, Q8_0, NVFP4 Mamba) + vision | 1x NVIDIA RTX 5090 (32 GB) | 150/150/150/150 | | 0.03 | tools 58/60, json 30/30, gsm8k 241/250, humaneval 153/164, ifeval 120/150 | 2026-10-15 |
| Qwen3.8-27B, MLX 4-bit + DFlash2 | 1x NVIDIA RTX 5090 (32 GB) | 241/459/310/314 | | 0.10 | | 2026-10-16 |
| Qwen3.8-Flash-Next INT4-Mixed (AutoRound) | 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE | 78/127/102/130 | 16: 304/516/392/525 | 0.03 / 0.45-0.54 | | 2026-10-17 |
| Qwen3.8-Flash-Next, 6-bit with vision | 1x Apple M5 Ultra (256 GB unified), macOS | 142/210/173/206 | 2: 198/262/222/272 | 0.11 / 0.17-0.19 | | 2026-10-17 |
| Qwen3.8-Flash-Next, MLX 6-bit | 1x Apple M5 Ultra (256 GB unified), macOS | 145/206/172/194 | 4: 566/816/673/854 | 0.09 / 1.63-3.07 | gsm8k 242/250, ifeval 134/150 | 2026-10-11 |
| Step-5-Preview, MLX 2-bit experts (8-bit rest) + MTP | 1x Apple M5 Ultra (256 GB unified), macOS | 38/50/40/48 | | 0.24 | | 2026-10-16 |

# DeepSeek-V4.1-Flash-1017, 2/3-bit experts on 1x Apple M5 Ultra (256 GB unified), macOS

Target deepseek-v41-m5: DeepSeek-V4.1-Flash-1017, 2/3-bit experts with images and DSpark drafts, TensorFold native Metal on one M5 Ultra. Engine: TensorFold native, Metal; served as DeepSeek-V4.1-Flash. Latest: run 20261020-043028-dsv41-speed (2026-10-20 04:30, plan dsv41-speed, done).

Serving: engine_commit bfad20ae; binary_sha256 9edbd25fa17c85acde00d3adddc35ae4d8f10cc487456a9c37261e840456cfb0; served_model {"object": "list", "data": [{"id": "DeepSeek-V4.1-Flash-Abliterated", "object": "model", "; checkpoint {'repo': 'deepseek-ai/DeepSeek-V4.1-Flash-1017', 'expert_bits': 2, 'layer_bits': '10-29:3'; chip Apple M5 Ultra; 274877906944; suite_commit 277c842a1481

## Ruler-streams

| streams | prose total (a stream) | code total (a stream) | structured total (a stream) | list total (a stream) | first token |
|---|---|---|---|---|---|
| 1 | 70.5 (70.5) | 88.5 (88.5) | 80.2 (80.2) | 93.6 (93.6) | 0.10-0.14 s |
| 2 | 88.8 (44.4) | 103.6 (51.8) | 96.2 (48.1) | 117.2 (58.6) | 0.21-0.28 s |
| 4 | 106.9 (26.5) | 120.2 (29.8) | 115.5 (29.2) | 124.0 (30.3) | 0.44-0.58 s |

## Checks

| check | result |
|---|---|
| memory-idle | measured: mac: System-wide memory free percentage: 14%; total = 5120.00M us |

## History

| run | plan | status | 1 stream code | 16 streams code | 16 streams first token | suite |
|---|---|---|---|---|---|---|
| 2026-10-20 04:30 | dsv41-speed | done | 88.5 | | | |
| 2026-10-19 09:08 | dsv41-speed | done | 90.5 | | | |
| 2026-10-18 20:28 | dsv41-quality | done | | | | 1487/1614 |
| 2026-10-18 18:44 | speed | done | 87.5 | | | |
| 2026-10-18 17:24 | speed | done | 92.5 | | | |
| 2026-10-17 19:33 | speed | done | 96.3 | | | |
| 2026-10-17 17:54 | speed | done | 90.5 | | | |
| 2026-10-17 14:57 | speed | done | 56.7 | | | |

# DeepSeek-V4.1-Flash-1017 EXL3 2.9bpw on 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over Ro

Target dsv41-1017-gb10-native-vision: DeepSeek-V4.1-Flash-1017 EXL3 2.9 bpw on native TensorFold (main + jayleaton's kit engine 9131e10a ported onto it), native images and video, TP=2 on two DGX Sparks, 4 slots, pool 200K. Engine: TensorFold native (Zig, CUDA): main 9ccf4727 + the DeepSeek-V4.1 kit engine, vision tower (TF_DSV41_IMAGES=native), video as sampled frames, DSpark drafts; served as DeepSeek-V4.1-Flash-TF. Latest: run 20261018-032401-dsv41-speed (2026-10-18 03:24, plan dsv41-speed, done).

Serving: image 7b3d42e0e5865d34; pack 778baeaf9fbcd351

## Ruler-streams

| streams | prose total (a stream) | code total (a stream) | structured total (a stream) | list total (a stream) | first token |
|---|---|---|---|---|---|
| 1 | 55.2 (55.2) | 100.4 (100.4) | 62.5 (62.5) | 71.4 (71.4) | 0.19-0.26 s |
| 2 | 81.0 (40.5) | 124.9 (62.5) | 84.2 (42.1) | 143.6 (71.8) | 0.31-0.40 s |
| 4 | 106.5 (26.4) | 161.6 (40.0) | 111.9 (26.9) | 178.7 (43.1) | 0.73-0.99 s |

## Checks

| check | result |
|---|---|
| memory-idle | measured: head 6.6 GiB available |

## History

| run | plan | status | 1 stream code | 16 streams code | 16 streams first token | suite |
|---|---|---|---|---|---|---|
| 2026-10-18 03:24 | dsv41-speed | done | 100.4 | | | |

# DeepSeek-V4.1-Flash-1017 EXL3 2.9bpw on 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over Ro

Target dsv41-1017-gb10-native-vision8: DeepSeek-V4.1-Flash-1017 EXL3 2.9 bpw on native TensorFold (main + the kit engine), native images and video, TP=2 on two DGX Sparks, 8 slots, pool 400K, prefill-dip reserve 2 GiB. Engine: TensorFold native (Zig, CUDA): main 9ccf4727 + the DeepSeek-V4.1 kit engine, vision tower, video as frames, DSpark drafts, 8 slots; served as DeepSeek-V4.1-Flash-TF. Latest: run 20261018-092218-dsv41-speed8 (2026-10-18 09:22, plan dsv41-speed8, done).

Serving: image 7b3d42e0e5865d34; pack 778baeaf9fbcd351

## Ruler-streams

| streams | prose total (a stream) | code total (a stream) | structured total (a stream) | list total (a stream) | first token |
|---|---|---|---|---|---|
| 1 | 55.4 (55.4) | 103.5 (103.5) | 62.7 (62.7) | 70.9 (70.9) | 0.19-0.26 s |
| 2 | 81.5 (40.7) | 126.2 (63.1) | 89.4 (44.7) | 142.7 (71.4) | 0.30-0.48 s |
| 4 | 108.8 (26.8) | 163.8 (40.3) | 113.7 (28.4) | 180.4 (44.4) | 0.71-0.85 s |
| 8 | 134.8 (16.7) | 202.7 (25.4) | 143.6 (17.4) | 203.3 (24.8) | 1.35-1.84 s |

## Checks

| check | result |
|---|---|
| memory-idle | measured: head 6.3 GiB available |

## History

| run | plan | status | 1 stream code | 16 streams code | 16 streams first token | suite |
|---|---|---|---|---|---|---|
| 2026-10-18 09:22 | dsv41-speed8 | done | 103.5 | | | |

# DeepSeek-V4.1-Flash-1017 EXL3 2.9bpw on 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over Ro

Target dsv41-1017-gb10-native: DeepSeek-V4.1-Flash-1017 EXL3 2.9 bpw on native TensorFold (main + the deepseek_v41 CUDA family, PRs 667-727), TP=2 on two DGX Sparks, 4 slots, pool 900K. Engine: TensorFold native (Zig, CUDA): main 9ccf4727 + deepseek_v41 series, kit serving profile, DSpark drafts, images as placeholder; served as DeepSeek-V4.1-Flash-TF. Latest: run 20261018-020350-dsv41-speed (2026-10-18 02:03, plan dsv41-speed, done).

Serving: image 00b298dda4527ad6; pack 778baeaf9fbcd351

## Ruler-streams

| streams | prose total (a stream) | code total (a stream) | structured total (a stream) | list total (a stream) | first token |
|---|---|---|---|---|---|
| 1 | 53.3 (53.3) | 98.1 (98.1) | 61.0 (61.0) | 69.3 (69.3) | 0.20-0.28 s |
| 2 | 78.4 (39.2) | 121.8 (60.9) | 82.5 (41.3) | 139.0 (69.5) | 0.30-0.42 s |
| 4 | 105.7 (26.2) | 158.5 (39.4) | 110.6 (26.5) | 176.4 (41.4) | 0.75-1.00 s |

## Checks

| check | result |
|---|---|
| memory-idle | measured: head 6.3 GiB available |

## History

| run | plan | status | 1 stream code | 16 streams code | 16 streams first token | suite |
|---|---|---|---|---|---|---|
| 2026-10-18 02:03 | dsv41-speed | done | 98.1 | | | |

# DeepSeek-V4.1-Flash-1017 EXL3 2.9bpw on 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over Ro

Target dsv41-1017-gb10: DeepSeek-V4.1-Flash-1017 EXL3 2.9 bpw, jayleaton's DeepSeek-V4.1 recipe (Python, TensorFold 0.6.0 + patches), TP=2 on two DGX Sparks, 4 slots. Engine: TensorFold 0.6.0 (Python) + jayleaton's DeepSeek-V4.1 patches, recipe defaults (replay prefill, DSpark drafts); served as DeepSeek-V4.1-Flash-TF. Latest: run 20261017-231853-dsv41-speed (2026-10-17 23:18, plan dsv41-speed, done).

Serving: image eb2f599de2fb618a; pack 778baeaf9fbcd351

## Ruler-streams

| streams | prose total (a stream) | code total (a stream) | structured total (a stream) | list total (a stream) | first token |
|---|---|---|---|---|---|
| 1 | 54.7 (54.7) | 100.6 (100.6) | 62.6 (62.6) | 71.3 (71.3) | 0.19-0.26 s |
| 2 | 77.4 (38.7) | 126.5 (63.3) | 84.0 (42.0) | 142.6 (71.3) | 0.30-0.39 s |
| 4 | 103.5 (25.4) | 155.8 (38.4) | 109.0 (27.7) | 173.2 (41.0) | 0.41-0.64 s |

## Checks

| check | result |
|---|---|
| memory-idle | measured: head 8.1 GiB available |

## History

| run | plan | status | 1 stream code | 16 streams code | 16 streams first token | suite |
|---|---|---|---|---|---|---|
| 2026-10-17 23:18 | dsv41-speed | done | 100.6 | | | |

# Gemma 4 12B, MLX 4-bit + MTP assistant on 1x Apple M5 Ultra (256 GB unified), macOS

Target gemma4-12b-m5: Gemma 4 12B 4-bit (groups of 64), MTP assistant drafts, images, TensorFold native Metal on one M5 Ultra. Engine: TensorFold native, Metal; served as gemma4-12b. Latest: run 20261016-113616-g12-quality (2026-10-16 11:36, plan g12-quality, done).

Serving: engine_commit 74005ebe; binary_sha256 215b4b7e3cd14a9d8178b4dc885d9f1dffbebd0f4ec2e9b274653820b71da8d2; served_model {"object": "list", "data": [{"id": "gemma4-12b", "object": "model", "created": 1792168585,; chip Apple M5 Ultra; 274877906944

## Suite

| check | passed |
|---|---|
| tools | 58/60 |
| json | 30/30 |
| gsm8k | 239/250 |
| mmlu | 264/342 |

## History

| run | plan | status | 1 stream code | 16 streams code | 16 streams first token | suite |
|---|---|---|---|---|---|---|
| 2026-10-16 11:36 | g12-quality | done | | | | 591/682 |
| 2026-10-16 11:31 | quick | done | 223.3 | | | 112/114 |

# Gemma 4 12B, oQ8e (8-bit) + MTP assistant on 1x Apple M5 Ultra (256 GB unified), macOS

Target gemma4-12b-oq8e-m5: Gemma 4 12B oQ8e (8-bit, groups of 64), MTP assistant drafts, images, TensorFold native Metal on one M5 Ultra. Engine: TensorFold native, Metal; served as gemma4-12b-oq8e. Latest: run 20261016-175459-speed (2026-10-16 17:54, plan speed, done).

Serving: engine_commit d9e36b7f; binary_sha256 2e29fee5da96ec1ffb1ba41302ac949dc12205572154f21b800823450eb4b2b0; served_model {"object": "list", "data": [{"id": "gemma4-12b-oq8e", "object": "model", "created": 179219; chip Apple M5 Ultra; 274877906944

## Ruler-fresh

| streams | prose total (a stream) | code total (a stream) | structured total (a stream) | list total (a stream) | first token |
|---|---|---|---|---|---|
| 1 | 92.8 (92.8) | 142.3 (142.3) | 145.7 (145.7) | 158.1 (158.1) | 0.05-0.05 s |
| 4 | 261.5 (64.8) | 324.2 (81.3) | 328.5 (80.7) | 340.0 (83.6) | 0.12-0.12 s |

Engine rounds (ruler-fresh):

| streams | tokens a round | drafts a round | accepted | ms a round | requests |
|---|---|---|---|---|---|
| 1 | 3.55 | 5.525 | 47% | 26.35 | 12 |
| 4 | 2.944 | 3.177 | 62% | 37.94 | 49 |

## Suite

| check | passed |
|---|---|
| drafts | 12/12 |
| concurrent | 12/12 |

## Checks

| check | result |
|---|---|
| structured | pass: 18/18 obey; a bad regex -> 400 |

## History

| run | plan | status | 1 stream code | 16 streams code | 16 streams first token | suite |
|---|---|---|---|---|---|---|
| 2026-10-16 17:54 | speed | done | 142.3 | | | 24/24 |
| 2026-10-16 17:37 | speed | done | 141.2 | | | 24/24 |

# Gemma 4 31B, MLX 4-bit + DFlash on 1x Apple M5 Ultra (256 GB unified), macOS

Target gemma4-31b-m5: Gemma 4 31B 4-bit (groups of 64), DFlash drafts, images, TensorFold native Metal on one M5 Ultra. Engine: TensorFold native, Metal; served as gemma4-31b. Latest: run 20261016-085420-quick (2026-10-16 08:54, plan quick, done).

Serving: engine_commit e8b60549; binary_sha256 d90e44c9818187ae4bc61d06312989cd796033b5dea6e0fc02164fe2730660fe; served_model {"object": "list", "data": [{"id": "gemma4-31b", "object": "model", "created": 1792158874,; chip Apple M5 Ultra; 274877906944

## Ruler-fresh

| streams | prose total (a stream) | code total (a stream) | structured total (a stream) | list total (a stream) | first token |
|---|---|---|---|---|---|
| 1 | 87.1 (87.1) | 157.9 (157.9) | 156.7 (156.7) | 130.4 (130.4) | 0.08-0.10 s |

Engine rounds (ruler-fresh):

| streams | tokens a round | drafts a round | accepted | ms a round | requests |
|---|---|---|---|---|---|
| 1 | 3.288 | 13.003 | 18% | 24.71 | 12 |

## Ruler-after

| streams | prose total (a stream) | code total (a stream) | structured total (a stream) | list total (a stream) | first token |
|---|---|---|---|---|---|
| 1 | 92.1 (92.1) | 161.4 (161.4) | 146.6 (146.6) | 126.1 (126.1) | 0.09-0.10 s |

Engine rounds (ruler-after):

| streams | tokens a round | drafts a round | accepted | ms a round | requests |
|---|---|---|---|---|---|
| 1 | 3.333 | 13.215 | 18% | 25.34 | 17 |

## Suite

| check | passed |
|---|---|
| soak | 60/60 |
| drafts | 12/12 |
| concurrent | 12/12 |
| streaming | 4/4 |
| multiturn | 8/8 |
| stop | 3/3 |

| check | passed |
|---|---|
| thinking | 10/10 |
| copy | 2/2 |
| max-tokens | 3/3 |

## Checks

| check | result |
|---|---|
| memory-idle | measured: mac: System-wide memory free percentage: 75%; total = 3072.00M us |
| structured | pass: 18/18 obey; a bad regex -> 400 |
| sampling | pass: 9/9 seeded replies equal on repeat and undrafted; 0 errors |
| turns | pass: resumed == again == cold: True; cached 5494 of 5564; ttft resumed 0.137s, cold 0.137s |

## History

| run | plan | status | 1 stream code | 16 streams code | 16 streams first token | suite |
|---|---|---|---|---|---|---|
| 2026-10-16 08:54 | quick | done | 157.9 | | | 114/114 |
| 2026-10-16 06:28 | quick | done | 156.5 | | | 114/114 |
| 2026-10-15 20:09 | quick | done | 153.8 | | | 114/114 |

# Gemma 4 31B, oQ8e (8-bit) + DFlash on 1x Apple M5 Ultra (256 GB unified), macOS

Target gemma4-31b-oq8e-m5: Gemma 4 31B oQ8e (8-bit, groups of 64), DFlash drafts, images, TensorFold native Metal on one M5 Ultra. Engine: TensorFold native, Metal; served as gemma4-31b-oq8e. Latest: run 20261016-180201-speed (2026-10-16 18:02, plan speed, done).

Serving: engine_commit d9e36b7f; binary_sha256 2e29fee5da96ec1ffb1ba41302ac949dc12205572154f21b800823450eb4b2b0; served_model {"object": "list", "data": [{"id": "gemma4-31b-oq8e", "object": "model", "created": 179219; chip Apple M5 Ultra; 274877906944

## Ruler-fresh

| streams | prose total (a stream) | code total (a stream) | structured total (a stream) | list total (a stream) | first token |
|---|---|---|---|---|---|
| 1 | 40.3 (40.3) | 63.1 (63.1) | 52.4 (52.4) | 54.8 (54.8) | 0.10-0.11 s |
| 4 | 109.7 (27.7) | 172.9 (44.8) | 141.5 (34.2) | 180.3 (44.3) | 0.24-0.27 s |

Engine rounds (ruler-fresh):

| streams | tokens a round | drafts a round | accepted | ms a round | requests |
|---|---|---|---|---|---|
| 1 | 2.759 | 10.847 | 16% | 52.41 | 12 |
| 4 | 2.384 | 3.645 | 38% | 63.19 | 49 |

## Suite

| check | passed |
|---|---|
| drafts | 12/12 |
| concurrent | 12/12 |

## Checks

| check | result |
|---|---|
| structured | pass: 18/18 obey; a bad regex -> 400 |

## History

| run | plan | status | 1 stream code | 16 streams code | 16 streams first token | suite |
|---|---|---|---|---|---|---|
| 2026-10-16 18:02 | speed | done | 63.1 | | | 24/24 |

# GLM-5.3-Flash EXL3 4bpw on 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE

Target glm53-exl3-gb10-best: GLM-5.3-Flash EXL3 4bpw, families/glm with 4-bit dense / 8-bit head, DFlash2 and 16 streams (PRs 855+979+984 and packing), TP=2 on two DGX Sparks. Engine: TensorFold native, CUDA: families/glm, packed linears (4/8, bf16 freed), DFlash2, 16 streams, prompts together (prefill_many); served as GLM-5.3-Flash-EXL3. Latest: run 20261017-195233-glm-ab-ifeval (2026-10-17 19:52, plan glm-ab-ifeval, done).

Serving: binary cf50867102a60c38

## Suite

| check | passed |
|---|---|
| ifeval | 103/150 |
| json | 30/30 |

## History

| run | plan | status | 1 stream code | 16 streams code | 16 streams first token | suite |
|---|---|---|---|---|---|---|
| 2026-10-17 19:52 | glm-ab-ifeval | done | | | | 133/180 |
| 2026-10-17 16:52 | glm-quality | done | 59.6 | | | 348/400 |
| 2026-10-17 16:26 | glm-streams | done | 59.5 | 131.3 | 1.27-1.66 s | 24/24 |
| 2026-10-17 15:56 | glm-streams | done | 67.7 | 132.8 | 1.23-1.59 s | 15/24 |
| 2026-10-17 15:28 | glm-streams | done | 68.3 | 127.5 | 2.63-3.30 s | 24/24 |
| 2026-10-17 14:38 | glm-streams | done | 68.7 | 123.7 | 1.77-2.24 s | 24/24 |
| 2026-10-17 12:40 | glm-streams | done | 69.4 | 125.0 | 2.55-3.17 s | 24/24 |
| 2026-10-17 11:31 | glm-streams | done | 69.8 | 136.4 | 2.55-3.14 s | 24/24 |

# GLM-5.3-Flash EXL3 4bpw on 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE

Target glm53-exl3-gb10-dflash2: GLM-5.3-Flash EXL3 4bpw, families/glm with DFlash2 drafts (--drafter, 4-bit), TP=2 on two DGX Sparks. Engine: TensorFold native, CUDA: families/glm + DFlash2 (incoai/GLM-5.3-Flash-DFlash2, opt-in); served as GLM-5.3-Flash-EXL3. Latest: run 20261017-003221-glm-streams8 (2026-10-17 00:32, plan glm-streams8, done).

Serving: binary 511e7e0521628504

## Ruler-streams

| streams | prose total (a stream) | code total (a stream) | structured total (a stream) | list total (a stream) | first token |
|---|---|---|---|---|---|
| 1 | 27.3 (27.3) | 48.4 (48.4) | 39.2 (39.2) | 33.1 (33.1) | 0.18-0.24 s |
| 4 | 60.0 (14.7) | 84.4 (20.8) | 75.6 (19.2) | 88.1 (21.6) | 0.71-0.88 s |
| 8 | 78.1 (9.7) | 115.4 (14.3) | 98.4 (12.4) | 109.8 (12.9) | 1.49-1.83 s |

## Suite-together

| check | passed |
|---|---|
| concurrent | 12/12 |
| drafts | 12/12 |

## History

| run | plan | status | 1 stream code | 16 streams code | 16 streams first token | suite |
|---|---|---|---|---|---|---|
| 2026-10-17 00:32 | glm-streams8 | done | 48.4 | | | 24/24 |

# GLM-5.3-Flash EXL3 4bpw on 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE

Target glm53-exl3-gb10-main: TensorFold main 76aafcf2 (with #855, #979, #984, #995 merged): GLM-5.3-Flash EXL3 4bpw, DFlash2, 16 streams, TP=2 on two DGX Sparks. Engine: TensorFold native, CUDA: families/glm (main); served as GLM-5.3-Flash-EXL3. Latest: run 20261018-131205-glm-qual1014 (2026-10-18 13:12, plan glm-qual1014, done).

Serving: binary e2169dae62202aec

## Ruler-c1-16

| streams | prose total (a stream) | code total (a stream) | structured total (a stream) | list total (a stream) | first token |
|---|---|---|---|---|---|
| 1 | 42.8 (42.3) | 67.2 (67.4) | 55.3 (54.8) | 53.1 (52.9) | 0.16-0.22 s |

## Ruler-streams

| streams | prose total (a stream) | code total (a stream) | structured total (a stream) | list total (a stream) | first token |
|---|---|---|---|---|---|
| 8 | 90.0 (11.2) | 120.8 (15.1) | 114.7 (14.1) | 115.5 (14.1) | 1.23-1.57 s |
| 16 | 107.2 (6.6) | 119.6 (7.3) | 119.4 (7.3) | 129.6 (8.1) | 2.69-3.38 s |

## Suite

| check | passed |
|---|---|
| concurrent | 12/12 |
| drafts | 12/12 |

## History

| run | plan | status | 1 stream code | 16 streams code | 16 streams first token | suite |
|---|---|---|---|---|---|---|
| 2026-10-18 13:12 | glm-qual1014 | done | | 119.6 | 2.69-3.38 s | 24/24 |

# GLM-5.3-Flash EXL3 4bpw on 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE

Target glm53-exl3-gb10-many-mtp: #1014 merged onto TensorFold main 76aafcf2, MTP head only (no --drafter), GLM-5.3-Flash EXL3 4bpw, TP=2. Engine: TensorFold native, CUDA: families/glm (many-mtp); served as GLM-5.3-Flash-EXL3. Latest: run 20261018-205115-glm-qual1014 (2026-10-18 20:51, plan glm-qual1014, done).

Serving: binary 4f1a478269c0e4fa

## Ruler-c1-16

| streams | prose total (a stream) | code total (a stream) | structured total (a stream) | list total (a stream) | first token |
|---|---|---|---|---|---|
| 1 | 42.1 (41.9) | 52.5 (52.2) | 50.2 (49.9) | 49.0 (49.2) | 0.17-0.22 s |

## Ruler-streams

| streams | prose total (a stream) | code total (a stream) | structured total (a stream) | list total (a stream) | first token |
|---|---|---|---|---|---|
| 8 | 74.8 (9.3) | 95.5 (11.9) | 100.9 (12.6) | 106.0 (13.2) | 0.84-1.05 s |
| 16 | 105.9 (6.6) | 119.7 (7.3) | 131.4 (8.3) | 138.3 (8.5) | 1.32-1.62 s |

## Suite

| check | passed |
|---|---|
| concurrent | 12/12 |
| drafts | 12/12 |

## History

| run | plan | status | 1 stream code | 16 streams code | 16 streams first token | suite |
|---|---|---|---|---|---|---|
| 2026-10-18 20:51 | glm-qual1014 | done | | 119.7 | 1.32-1.62 s | 24/24 |
| 2026-10-18 14:04 | glm-qual1014 | done | | 118.6 | 1.28-1.58 s | 24/24 |

# GLM-5.3-Flash EXL3 4bpw on 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE

Target glm53-exl3-gb10-many: #1014 merged onto TensorFold main 76aafcf2: batched prompts (prefill_many) with DFlash2's feed and the hold fix, GLM-5.3-Flash EXL3 4bpw, TP=2. Engine: TensorFold native, CUDA: families/glm (many); served as GLM-5.3-Flash-EXL3. Latest: run 20261018-202804-glm-qual1014 (2026-10-18 20:28, plan glm-qual1014, done).

Serving: binary 4f1a478269c0e4fa

## Ruler-c1-16

| streams | prose total (a stream) | code total (a stream) | structured total (a stream) | list total (a stream) | first token |
|---|---|---|---|---|---|
| 1 | 42.5 (42.6) | 67.6 (68.2) | 56.1 (55.9) | 55.8 (53.2) | 0.17-0.23 s |

## Ruler-streams

| streams | prose total (a stream) | code total (a stream) | structured total (a stream) | list total (a stream) | first token |
|---|---|---|---|---|---|
| 8 | 92.4 (11.6) | 131.0 (16.2) | 125.4 (15.6) | 147.8 (17.1) | 0.82-1.03 s |
| 16 | 117.4 (7.3) | 134.1 (8.1) | 147.9 (9.3) | 158.6 (9.7) | 1.34-1.66 s |

## Suite

| check | passed |
|---|---|
| concurrent | 12/12 |
| drafts | 12/12 |

## History

| run | plan | status | 1 stream code | 16 streams code | 16 streams first token | suite |
|---|---|---|---|---|---|---|
| 2026-10-18 20:28 | glm-qual1014 | done | | 134.1 | 1.34-1.66 s | 24/24 |
| 2026-10-18 14:29 | glm-quality | done | 60.1 | | | 1947/2418 |
| 2026-10-18 13:40 | glm-qual1014 | done | | 134.1 | 1.35-1.67 s | 24/24 |

# GLM-5.3-Flash EXL3 4bpw on 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE

Target glm53-exl3-gb10-packed: GLM-5.3-Flash EXL3 4bpw, families/glm with 4-bit dense / 8-bit head, DFlash2 and 16 streams (PRs 855+979+984 and packing), TP=2 on two DGX Sparks. Engine: TensorFold native, CUDA: families/glm, packed linears (4/8, bf16 freed), DFlash2, 16 streams, prompts one at a time; served as GLM-5.3-Flash-EXL3. Latest: run 20261017-195233-glm-ab-ifeval (2026-10-17 19:52, plan glm-ab-ifeval, done).

Serving: binary cf50867102a60c38

## Suite

| check | passed |
|---|---|
| ifeval | 103/150 |
| json | 30/30 |

## History

| run | plan | status | 1 stream code | 16 streams code | 16 streams first token | suite |
|---|---|---|---|---|---|---|
| 2026-10-17 19:52 | glm-ab-ifeval | done | | | | 133/180 |

# GLM-5.3-Flash EXL3 4bpw on 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE

Target glm53-exl3-gb10-port: GLM-5.3-Flash EXL3 4bpw, main's families/glm with PR 855's port, TP=2 on two DGX Sparks. Engine: TensorFold native, CUDA: families/glm on 1.0.12 (PR 855's port); served as GLM-5.3-Flash-EXL3. Latest: run 20261016-213802-glm-streams (2026-10-16 21:38, plan glm-streams, done).

Serving: binary 2b666cc74334abec

## Ruler-streams

| streams | prose total (a stream) | code total (a stream) | structured total (a stream) | list total (a stream) | first token |
|---|---|---|---|---|---|
| 1 | 27.9 (27.9) | 38.0 (38.0) | 33.2 (33.2) | 30.6 (30.6) | 0.18-0.23 s |
| 4 | 48.9 (12.3) | 66.0 (16.7) | 64.8 (16.2) | 63.7 (16.1) | 0.79-0.86 s |
| 8 | 60.3 (7.6) | 75.9 (9.5) | 82.2 (10.3) | 76.3 (9.4) | 1.48-1.73 s |
| 16 | 123.6 (7.6) | 155.5 (9.6) | 149.5 (9.0) | 145.0 (9.1) | 8.50-17.20 s |

## Suite-together

| check | passed |
|---|---|
| concurrent | 12/12 |
| drafts | 12/12 |

## History

| run | plan | status | 1 stream code | 16 streams code | 16 streams first token | suite |
|---|---|---|---|---|---|---|
| 2026-10-16 21:38 | glm-streams | done | 38.0 | 155.5 | 8.50-17.20 s | 24/24 |
| 2026-10-16 21:24 | glm-vision | done | | | | 22/22 |
| 2026-10-16 21:03 | glm-vision | done | | | | 4/6 |
| 2026-10-16 18:10 | glm-quality | done | 38.2 | | | |

# GLM-5.3-Flash EXL3 4bpw on 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE

Target glm53-exl3-gb10-port16: GLM-5.3-Flash EXL3 4bpw, families/glm with 4096-row index rings, --parallel 16, TP=2 on two DGX Sparks. Engine: TensorFold native, CUDA: families/glm + index rings (16 streams at 40960); served as GLM-5.3-Flash-EXL3. Latest: run 20261017-062916-glm-streams (2026-10-17 06:29, plan glm-streams, done).

Serving: binary 142bc21ac97bc065

## Ruler-streams

| streams | prose total (a stream) | code total (a stream) | structured total (a stream) | list total (a stream) | first token |
|---|---|---|---|---|---|
| 1 | 28.0 (28.0) | 38.6 (38.6) | 33.5 (33.5) | 30.7 (30.7) | 0.18-0.23 s |
| 4 | 51.3 (12.9) | 65.5 (16.3) | 65.7 (16.5) | 66.0 (16.5) | 0.67-0.81 s |
| 8 | 64.3 (8.1) | 82.3 (10.3) | 83.2 (10.4) | 81.8 (10.2) | 1.41-1.68 s |
| 16 | 86.4 (5.4) | 100.6 (6.3) | 100.2 (6.1) | 101.5 (6.2) | 3.23-3.72 s |

## Suite-together

| check | passed |
|---|---|
| concurrent | 12/12 |
| drafts | 12/12 |

## History

| run | plan | status | 1 stream code | 16 streams code | 16 streams first token | suite |
|---|---|---|---|---|---|---|
| 2026-10-17 06:29 | glm-streams | done | 38.6 | 100.6 | 3.23-3.72 s | 24/24 |
| 2026-10-17 05:56 | glm-streams | done | 39.0 | 90.8 | 3.20-3.67 s | 24/24 |

# GLM-5.3-Flash, EXL3 4 bpw on 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE

Target glm53-exl3-sparks: GLM-5.3-Flash EXL3 4 bpw, TensorFold native, TP=2 on two DGX Sparks. Engine: TensorFold native, CUDA; served as GLM-5.3-Flash-EXL3. Latest: run evidence/2026-10-14 (2026-10-14 00:00, plan published evidence, published).

### Ruler-shipped-14fb445e-c1-sampled

| streams | prose total (a stream) | code total (a stream) | structured total (a stream) | list total (a stream) | first token |
|---|---|---|---|---|---|
| 1 | 45.1 (45.2) | 64.6 (65.4) | 56.5 (56.2) | 57.0 (54.9) | 0.18-0.25 s |

### Ruler-shipped-14fb445e-c1

| streams | prose total (a stream) | code total (a stream) | structured total (a stream) | list total (a stream) | first token |
|---|---|---|---|---|---|
| 1 | 46.5 (45.4) | 68.7 (69.4) | 57.0 (56.4) | 58.8 (55.3) | 0.18-0.24 s |

### Ruler-shipped-14fb445e-c16

| streams | prose total (a stream) | code total (a stream) | structured total (a stream) | list total (a stream) | first token |
|---|---|---|---|---|---|
| 16 | 144.0 (8.9) | 198.3 (12.1) | 180.1 (11.3) | 208.4 (12.5) | 0.91-1.19 s |

### Ruler-shipped-14fb445e-c8

| streams | prose total (a stream) | code total (a stream) | structured total (a stream) | list total (a stream) | first token |
|---|---|---|---|---|---|
| 8 | 110.1 (13.6) | 152.5 (18.6) | 138.7 (17.5) | 157.0 (18.7) | 0.49-0.63 s |

### Ruler-shipped-f95f2f70-c1-sampled

| streams | prose total (a stream) | code total (a stream) | structured total (a stream) | list total (a stream) | first token |
|---|---|---|---|---|---|
| 1 | 45.4 (45.4) | 65.8 (67.0) | 56.4 (55.9) | 58.1 (55.3) | 0.17-0.23 s |

### Ruler-shipped-f95f2f70-c1

| streams | prose total (a stream) | code total (a stream) | structured total (a stream) | list total (a stream) | first token |
|---|---|---|---|---|---|
| 1 | 47.5 (46.5) | 69.5 (69.9) | 58.2 (56.9) | 60.2 (56.8) | 0.16-0.23 s |

## Ruler-shipped-f95f2f70-c16

| streams | prose total (a stream) | code total (a stream) | structured total (a stream) | list total (a stream) | first token |
|---|---|---|---|---|---|
| 16 | 144.8 (8.9) | 202.2 (12.3) | 181.4 (11.4) | 207.6 (12.5) | 0.85-1.14 s |

## Ruler-shipped-f95f2f70-c8

| streams | prose total (a stream) | code total (a stream) | structured total (a stream) | list total (a stream) | first token |
|---|---|---|---|---|---|
| 8 | 111.1 (13.7) | 154.1 (18.7) | 139.5 (17.2) | 157.3 (18.8) | 0.45-0.60 s |

## Ablit/suite

| check | passed |
|---|---|
| gsm8k | 246/250 |
| mmlu | 292/342 |
| humaneval | 159/164 |
| ifeval | 118/150 |
| tools | 59/60 |
| json | 30/30 |
| mgsm | 221/240 |

## Suite-ab

| check | passed |
|---|---|
| gsm8k | 244/250 |
| mmlu | 292/342 |
| humaneval | 156/164 |
| ifeval | 118/150 |
| tools | 59/60 |
| json | 30/30 |
| mgsm | 223/240 |

## History

| run | plan | status | 1 stream code | 16 streams code | 16 streams first token | suite |
|---|---|---|---|---|---|---|
| 2026-10-14 00:00 | published evidence | published | 68.7 | | | 1125/1236 |
| 2026-10-13 00:00 | published evidence | published | 1083.6 | | | 1874/2172 |
| 2026-10-12 00:00 | published evidence | published | 62.8 | 169.5 | 0.45-1.19 s | |

# GLM-5.3-Flash, MLX 4-bit on 1x Apple M5 Ultra (256 GB unified), macOS

Target glm53-m5: GLM-5.3-Flash MLX 4-bit, TensorFold native on one M5 Ultra Mac (one stream, greedy). Engine: TensorFold native, Metal; served as GLM-5.3-Flash-MLX-4bit. Latest: run 20261016-082326-glm-gate-novision (2026-10-16 08:23, plan glm-gate-novision, done).

Serving: engine_commit e8b60549; binary_sha256 d90e44c9818187ae4bc61d06312989cd796033b5dea6e0fc02164fe2730660fe

## Ruler-fresh

| streams | prose total (a stream) | code total (a stream) | structured total (a stream) | list total (a stream) | first token |
|---|---|---|---|---|---|
| 1 | 74.4 (74.4) | 78.3 (78.3) | 77.5 (77.5) | 86.5 (86.5) | 0.11-0.15 s |

## Suite

| check | passed |
|---|---|
| drafts | 12/12 |
| concurrent | 12/12 |
| streaming | 4/4 |
| multiturn | 8/8 |
| stop | 3/3 |
| thinking | 10/10 |
| copy | 2/2 |
| max-tokens | 3/3 |
| tools | 57/60 |
| json | 30/30 |
| gsm8k | 246/250 |
| mmlu | 290/342 |

## History

| run | plan | status | 1 stream code | 16 streams code | 16 streams first token | suite |
|---|---|---|---|---|---|---|
| 2026-10-16 08:23 | glm-gate-novision | done | 78.3 | | | 677/736 |
| 2026-10-16 08:04 | glm-gate-novision | done | 77.6 | | | 679/736 |
| 2026-10-16 00:35 | glm-gate | done | 78.7 | | | 689/748 |
| 2026-10-10 00:00 | published evidence | published | 103.7 | | | 36/36 |

| run | plan | status | 1 stream code | 16 streams code | 16 streams first token | suite |
|---|---|---|---|---|---|---|
| 2026-10-10 00:00 | published evidence | published | | | | 1128/1213 |
| 2026-10-10 00:00 | published evidence | published | | | | 1137/1213 |
| 2026-10-10 00:00 | published evidence | published | | | | 1079/1213 |
| 2026-10-09 00:00 | published evidence | published | 104.6 | | | |
| 2026-10-09 00:00 | published evidence | published | | | | 349/403 |
| 2026-10-09 00:00 | published evidence | published | | | | 42/42 |

# NVIDIA Nemotron 3.5 Bolt 74B-A9B (24 GB: IQ2_XXS experts, Q8_0, NVFP4 Mamba) + vision on 1x NVIDIA RT

Target nemotron-bolt-5090: Nemotron 3.5 Bolt 74B-A9B ModelOpt IQ2_XXS/Q8_0/NVFP4, TensorFold native CUDA on one RTX 5090 (262144, two streams, vision, host prompt cache). Engine: TensorFold native, CUDA; served as nemotron-bolt. Latest: run 20261015-175621-full (2026-10-15 17:56, plan full, done).

Serving: recipe_commit f546a71; engine_commit b2ec378d; served_model {"object": "list", "data": [{"id": "nemotron-bolt", "object": "model", "created": 17920869

## Ruler-fresh

| streams | prose total (a stream) | code total (a stream) | structured total (a stream) | list total (a stream) | first token |
|---|---|---|---|---|---|
| 1 | 149.7 (149.7) | 149.6 (149.6) | 150.1 (150.1) | 149.7 (149.7) | 0.03-0.03 s |

Engine rounds (ruler-fresh):

| streams | tokens a round | drafts a round | accepted | ms a round | requests |
|---|---|---|---|---|---|
| 1 | 1.011 | 0.0 | | 6.75 | 217 |

## Ruler-after

| streams | prose total (a stream) | code total (a stream) | structured total (a stream) | list total (a stream) | first token |
|---|---|---|---|---|---|
| 1 | 149.3 (149.3) | 149.2 (149.2) | 149.7 (149.7) | 149.3 (149.3) | 0.03-0.03 s |

Engine rounds (ruler-after):

| streams | tokens a round | drafts a round | accepted | ms a round | requests |
|---|---|---|---|---|---|
| 1 | 1.007 | 0.0 | | 6.74 | 1781 |

## Suite

| check | passed |
|---|---|
| drafts | 12/12 |
| concurrent | 12/12 |
| streaming | 4/4 |
| multiturn | 8/8 |
| stop | 3/3 |
| thinking | 10/10 |
| copy | 2/2 |

| check | passed |
|---|---|
| max-tokens | 3/3 |
| tools | 58/60 |
| json | 30/30 |
| ifeval | 120/150 |
| gsm8k | 241/250 |
| mgsm | 190/240 |
| mmlu | 279/342 |
| humaneval | 153/164 |
| repetition | 8/8 |
| long-context | 55/55 |
| vision | 12/12 |
| soak | 60/60 |
| snapshot | 12/12 |

## Checks

| check | result |
|---|---|
| memory-idle | measured: gpu: 28716 MiB, 32607 MiB |
| structured | pass: 18/18 obey; a bad regex -> 400 |
| sampling | pass: 9/9 seeded replies equal on repeat; 0 errors |
| turns | pass: resumed == again == cold: True; cached 5832 of 5865; ttft resumed 0.053s, cold 0.134s |
| capacity | pass: 6/6 answered, 0 refused, 0 lost (no reply); a short request after: 200 |

## History

| run | plan | status | 1 stream code | 16 streams code | 16 streams first token | suite |
|---|---|---|---|---|---|---|
| 2026-10-15 17:56 | full | done | 149.6 | | | 1272/1437 |

# Qwen3.8-27B, MLX 4-bit + DFlash2 on 1x NVIDIA RTX 5090 (32 GB)

Target qwen38-27b-5090: Qwen3.8-27B MLX 4-bit + DFlash2, TensorFold native CUDA on one RTX 5090 (fp8 KV, 262k, one stream, vision). Engine: TensorFold native, CUDA; served as qwen38-27b. Latest: run 20261016-111449-quick (2026-10-16 11:14, plan quick, done).

Serving: recipe_commit 160a2c0; engine_commit bfcc84ab; served_model {"object": "list", "data": [{"id": "qwen38-27b", "object": "model", "created": 1792149307,

## Ruler-fresh

| streams | prose total (a stream) | code total (a stream) | structured total (a stream) | list total (a stream) | first token |
|---|---|---|---|---|---|
| 1 | 240.7 (240.7) | 459.0 (459.0) | 309.5 (309.5) | 313.5 (313.5) | 0.10-0.10 s |

## Ruler-after

| streams | prose total (a stream) | code total (a stream) | structured total (a stream) | list total (a stream) | first token |
|---|---|---|---|---|---|
| 1 | 239.8 (239.8) | 457.6 (457.6) | 308.9 (308.9) | 313.4 (313.4) | 0.10-0.10 s |

## Suite

| check | passed |
|---|---|
| soak | 60/60 |
| drafts | 12/12 |
| concurrent | 12/12 |
| streaming | 4/4 |
| multiturn | 7/8 |
| stop | 3/3 |
| thinking | 10/10 |
| copy | 2/2 |
| max-tokens | 3/3 |

## Checks

| check | result |
|---|---|
| memory-idle | measured: gpu: 20958 MiB, 32607 MiB |
| sampling | pass: 9/9 seeded replies equal on repeat and undrafted; 0 errors |

## History

| run | plan | status | 1 stream code | 16 streams code | 16 streams first token | suite |
|---|---|---|---|---|---|---|
| 2026-10-16 11:14 | quick | done | 459.0 | | | 113/114 |
| 2026-10-14 00:00 | published evidence | published | | | | |
| 2026-10-13 00:00 | published evidence | published | | | | |
| 2026-10-12 00:00 | published evidence | published | | | | 1229/1437 |
| 2026-10-12 00:00 | published evidence | published | | | | 1289/1437 |
| 2026-10-12 00:00 | published evidence | published | | | | |
| 2026-10-12 00:00 | published evidence | published | | | | |

# Qwen3.8-Flash-Next, MLX 6-bit on 1x Apple M5 Ultra (256 GB unified), macOS

Target qwen38-flashnext-mlx6bit-m5: Qwen3.8-Flash-Next MLX 6-bit, TensorFold native on one M5 Ultra. Engine: TensorFold native, Metal; served as Qwen3.8-Flash-Next-MLX-6bit. Latest: run evidence/2026-10-11-suite-rest (2026-10-11 00:00, plan published evidence, published).

## Suite

| check | passed |
|---|---|
| ifeval | 134/150 |
| gsm8k | 242/250 |
| mgsm | 225/240 |
| mmlu | 282/342 |
| repetition | 8/8 |
| long-context | 55/55 |
| concurrent | 12/12 |
| snapshot | 12/12 |

## History

| run | plan | status | 1 stream code | 16 streams code | 16 streams first token | suite |
|---|---|---|---|---|---|---|
| 2026-10-11 00:00 | published evidence | published | | | | 970/1069 |
| 2026-10-11 00:00 | published evidence | published | 205.9 | | | 24/24 |
| 2026-10-11 00:00 | published evidence | published | | | | 157/164 |
| 2026-10-11 00:00 | published evidence | published | 199.8 | | | |
| 2026-10-11 00:00 | published evidence | published | | | | 131/132 |
| 2026-10-11 00:00 | published evidence | published | 206.2 | | | |

# Qwen3.8-Flash-Next, 6-bit with vision on 1x Apple M5 Ultra (256 GB unified), macOS

Target qwen38-flashnext-vision-m5: Qwen3.8-Flash-Next 6-bit MTP with images and video, TensorFold native Metal on one M5 Ultra. Engine: TensorFold native, Metal; served as Qwen3.8-Flash-Next-MLX-6bit. Latest: run 20261017-120921-fn-speed (2026-10-17 12:09, plan fn-speed, done).

Serving: engine_commit 394d2e35; binary_sha256 43a9c01ac3eee31998def08595f5e4fc6d29819d46799a8ac01b7cdffd72b668; served_model {"object": "list", "data": [{"id": "Qwen3.8-Flash-Next-MLX-6bit", "object": "model", "crea; chip Apple M5 Ultra; 274877906944

## Ruler-fresh

| streams | prose total (a stream) | code total (a stream) | structured total (a stream) | list total (a stream) | first token |
|---|---|---|---|---|---|
| 1 | 142.4 (142.4) | 209.5 (209.5) | 173.2 (173.2) | 206.5 (206.5) | 0.11-0.13 s |
| 2 | 197.6 (98.8) | 261.9 (130.9) | 221.5 (110.7) | 271.5 (135.8) | 0.17-0.19 s |

## Suite

| check | passed |
|---|---|
| drafts | 12/12 |
| concurrent | 12/12 |

## Checks

| check | result |
|---|---|
| structured | pass: 18/18 obey; a bad regex -> 400 |

## History

| run | plan | status | 1 stream code | 16 streams code | 16 streams first token | suite |
|---|---|---|---|---|---|---|
| 2026-10-17 12:09 | fn-speed | done | 209.5 | | | 24/24 |
| 2026-10-17 12:06 | fn-speed | done | 209.8 | | | 24/24 |
| 2026-10-17 11:34 | fn-1m | done | 197.4 | | | 24/24 |
| 2026-10-17 11:31 | fn-media | done | 208.9 | | | 24/24 |
| 2026-10-17 08:04 | fn-speed | done | 206.1 | | | 24/24 |
| 2026-10-17 08:02 | fn-speed | done | 204.3 | | | 24/24 |
| 2026-10-17 05:58 | fn-speed | done | 208.0 | | | 24/24 |
| 2026-10-17 01:00 | fn-speed | done | 202.8 | | | 24/24 |

| run | plan | status | 1 stream code | 16 streams code | 16 streams first token | suite |
|---|---|---|---|---|---|---|
| 2026-10-17 00:57 | fn-speed | done | 207.3 | | | 24/24 |
| 2026-10-16 19:29 | fn-speed | done | 201.8 | | | 24/24 |
| 2026-10-16 19:26 | fn-speed | done | 207.3 | | | 24/24 |
| 2026-10-12 00:00 | published evidence | published | 174.1 | | | |
| 2026-10-12 00:00 | published evidence | published | 202.7 | | | |
| 2026-10-12 00:00 | published evidence | published | 176.6 | | | |
| 2026-10-11 00:00 | published evidence | published | 205.6 | | | 131/132 |
| 2026-10-11 00:00 | published evidence | published | 178.9 | | | 513/544 |

# Qwen3.8-Flash-Next INT4-Mixed (AutoRound) on 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 ov

Target qwen38-int4mixed: Qwen3.8-Flash-Next INT4-Mixed, TensorFold native, TP=2 on two DGX Sparks. Engine: TensorFold native, CUDA; served as Qwen3.8-Flash-Next-INT4-Mixed. Latest: run 20261017-072502-rebase (2026-10-17 07:25, plan rebase, done).

Serving: patch_sha 47c78d3f99593702c9e8a0baa64ec32c937520930186c78549004f411cd5f731; image tf-qwen38-int4mixed:native-v1.0.12 sha256:276fe2ba4ed3e778ec6f55ffad1430897b9ff439aff8352b; server_args --context 1048576 --no-thinking --parallel 16 --vision-urls --vision

## Ruler-fresh

| streams | prose total (a stream) | code total (a stream) | structured total (a stream) | list total (a stream) | first token |
|---|---|---|---|---|---|
| 1 | 78.0 (78.0) | 127.0 (127.0) | 102.1 (102.1) | 130.5 (130.5) | 0.03-0.04 s |
| 16 | 304.2 (18.9) | 516.3 (31.0) | 391.6 (24.0) | 524.9 (31.7) | 0.45-0.54 s |

Engine rounds (ruler-fresh):

| streams | tokens a round | drafts a round | accepted | ms a round | requests |
|---|---|---|---|---|---|
| 1 | 3.621 | 4.042 | 66% | 33.09 | 12 |
| 16 | 3.212 | 2.921 | 76% | 121.71 | 193 |

## Suite

| check | passed |
|---|---|
| soak | 60/60 |
| drafts | 12/12 |
| concurrent | 12/12 |
| streaming | 4/4 |
| multiturn | 8/8 |
| stop | 3/3 |
| thinking | 10/10 |
| copy | 2/2 |
| max-tokens | 3/3 |

## Suite-guided

| check | passed |
|---|---|
| json-guided | 30/30 |
| tools | 60/60 |

## Suite-thinking

| check | passed |
|---|---|
| gsm8k | 234/250 |

## Checks

| check | result |
|---|---|
| memory-idle | measured: head 22.6 GiB available; worker 25.3 GiB available |
| structured | pass: 18/18 obey; a bad regex -> 400 |
| sampling | pass: 9/9 seeded replies equal on repeat and undrafted; 0 errors |
| turns | pass: resumed == again == cold: True; cached 5575 of 5642; ttft resumed 0.105s, cold 0.035s |
| thinking | pass: thinking on: 10/10 right, 10/10 with reasoning_content, 0 leaks, 0 errors |

## History

| run | plan | status | 1 stream code | 16 streams code | 16 streams first token | suite |
|---|---|---|---|---|---|---|
| 2026-10-17 07:25 | rebase | done | 127.0 | 516.3 | 0.45-0.54 s | 114/114 |
| 2026-10-15 17:13 | full | done | 127.0 | 509.8 | 0.44-0.55 s | 1321/1437 |
| 2026-10-15 00:00 | published evidence | published | 123.4 | 472.9 | 0.52-0.74 s | 1321/1437 |
| 2026-10-15 00:00 | published evidence | published | | | | 1321/1437 |
| 2026-10-11 00:00 | published evidence | published | 120.1 | 507.2 | 0.54-0.62 s | 1321/1437 |
| 2026-10-11 00:00 | published evidence | published | 120.4 | 507.1 | 0.50-0.59 s | |
| 2026-10-10 09:45 | quick | done | 105.3 | | | 54/54 |
| 2026-10-10 00:00 | published evidence | published | 122.0 | 510.3 | 0.40-0.49 s | 114/114 |
| 2026-10-10 00:00 | published evidence | published | 125.9 | 510.9 | 0.44-0.49 s | 1261/1377 |
| 2026-10-10 00:00 | published evidence | published | 125.7 | 506.7 | 0.42-0.49 s | 54/54 |
| 2026-10-10 00:00 | published evidence | published | 106.1 | 506.1 | 0.42-0.51 s | 54/54 |

| run | plan | status | 1 stream code | 16 streams code | 16 streams first token | suite |
|---|---|---|---|---|---|---|
| 2026-10-10 00:00 | published evidence | published | 110.0 | 511.8 | 0.41-0.55 s | 1261/1377 |
| 2026-10-10 00:00 | published evidence | published | 105.8 | 505.5 | 0.41-0.46 s | 1261/1377 |
| 2026-10-09 00:00 | published evidence | published | 105.2 | 412.2 | 0.78-0.98 s | 1260/1377 |
| 2026-10-09 00:00 | published evidence | published | | | | 1256/1377 |
| 2026-10-08 00:00 | published evidence | published | 107.1 | 519.4 | 0.44-0.56 s | |
| 2026-10-08 00:00 | published evidence | published | 107.7 | 474.0 | 0.52-0.66 s | 145/145 |
| 2026-10-08 00:00 | published evidence | published | 107.6 | 442.6 | 0.44-0.58 s | 36/36 |
| 2026-10-08 00:00 | published evidence | published | 107.6 | 441.7 | 0.45-0.58 s | |
| 2026-10-07 00:00 | published evidence | published | | | | 12/12 |

# Step-5-Preview, MLX 2-bit experts (8-bit rest) + MTP on 1x Apple M5 Ultra (256 GB unified), macOS

Target step5-m5: Step-5-Preview 2-bit (experts 2/128, rest 8/64, MTP, images), TensorFold native Metal on one M5 Ultra. Engine: TensorFold native, Metal; served as step5-preview. Latest: run 20261016-102144-pr903-final (2026-10-16 10:21, plan pr903-final, done).

Serving: engine_commit e2bb58db; binary_sha256 61059a383f74fd363f17527526fdc7de028fe624d40e59e3238d20e4925bfef5; served_model {"object": "list", "data": [{"id": "step5-preview", "object": "model", "created": 17921641; chip Apple M5 Ultra; 274877906944

## Ruler-fresh

| streams | prose total (a stream) | code total (a stream) | structured total (a stream) | list total (a stream) | first token |
|---|---|---|---|---|---|
| 1 | 38.5 (38.5) | 50.4 (50.4) | 40.5 (40.5) | 47.9 (47.9) | 0.24-0.28 s |

Engine rounds (ruler-fresh):

| streams | tokens a round | drafts a round | accepted | ms a round | requests |
|---|---|---|---|---|---|
| 1 | 2.381 | 2.072 | 67% | 53.72 | 12 |

## Suite

| check | passed |
|---|---|
| vision | 12/12 |
| drafts | 12/12 |
| concurrent | 12/12 |

## Checks

| check | result |
|---|---|
| memory-idle | measured: mac: System-wide memory free percentage: 27%; total = 4096.00M us |
| turns | pass: resumed == again == cold: True; cached 4514 of 4618; ttft resumed 0.353s, cold 3.21s |

## History

| run | plan | status | 1 stream code | 16 streams code | 16 streams first token | suite |
|---|---|---|---|---|---|---|
| 2026-10-16 10:21 | pr903-final | done | 50.4 | | | 36/36 |
| 2026-10-16 09:23 | pr903 | done | 50.4 | | | 156/156 |
| 2026-10-15 19:44 | one-stream | done | 50.9 | | | 220/232 |

| run | plan | status | 1 stream code | 16 streams code | 16 streams first token | suite |
|---|---|---|---|---|---|---|
| 2026-10-15 19:28 | quick | done | 50.5 | | | 114/114 |
| 2026-10-15 18:52 | quick | done | 50.6 | | | 114/114 |
| 2026-10-15 13:43 | one-stream | done | 50.6 | | | 220/232 |
| 2026-10-15 12:43 | quick | done | 50.6 | | | |

# Served earlier, numbers in their READMEs only

These recipes predate the ruler's JSON, so their measurements are not in the ledger; their READMEs hold them.

| model | hardware | engine | recipe |
|---|---|---|---|
| Qwen3.8-Flash-Next, turboderp EXL3 4.05 bpw | 2x NVIDIA DGX Spark (GB10), TP=2 | TensorFold 0.6.6 (Python) with an EXL3 patch | github.com/BobClawblaw/qwen38-flashnext-exl3-2x-dgx-sparks |
| Qwen3.8-Flash-Next (TensorFold 0.6.6 recipe) | 2x NVIDIA DGX Spark (GB10), TP=2 | TensorFold 0.6.6 (Python) | github.com/BobClawblaw/qwen38-flashnext-tensorfold-2x-dgx-sparks |

# Step-5-Preview on one M5 Ultra, final build: measured outside the harness

TensorFold native Metal, branch step5-native at engine 6e791da6: 2,048-row prompt chunks, the prompt cache and the idle keepalive's kernel tick. Greedy unless said; served with --parallel 2 --vision at the default context (227,025).

| what | measured |
|---|---|
| prompt, 4k / 8k / 16k / 32k tokens | 868 / 863 / 840 / 800 tok/s (a 4k prompt right after a pause was 580 before the keepalive fix) |
| 212k-token chat question, one record's code of 10,500 | 2 of 3 right (records 3,000 and 9,800; missed 777), 376 s each, at least 19% of memory free |
| 129k-token chat question | 2 of 2 right (records 777 and 5,000), about 200 s each |
| MTP drafts against "draft": false, 200-400-token replies timed with their prompt | identical; code 25.8 to 41.2 tok/s, prose 27.0 to 36.6, math 26.9 to 48.8 |
| prompt cache, an 11.6k-token chat sent again, then two more turns | 15.3 s fresh, 1.8 s again, turns resumed at 11,601; replies identical to --prompt-cache-gib 0 |
| prompt cache, images (1200 x 800 PNG, 688 tokens) | resent in 2.15 s instead of 3.8 s; replies identical to the cache off |
| a kept state that cannot sit beside its stream (2.4 GiB cache budget) | dropped before the pass; the prompt read again with the same reply |
| a 1-token reply after 2 / 6 / 30 / 120 s idle | 0.17 / 0.18 / 0.18 / 0.19 s (about 3 s before the kernel tick and residency set) |
| two 18k/31k-token chats at once | both right |

# Step-5-Preview on one M5 Ultra, first build (512-row chunks): measured outside the harness

TensorFold native Metal, branch step5-native (engine 5ad7660c, images aa221b11), MLX 2-bit experts with the rest 8-bit, greedy, one stream. The harness's step5-m5 runs hold the ruler and the suite; these are the rest.

| what | measured |
|---|---|
| prompt, 4k / 8k / 16k / 32k tokens | 690 / 668 / 647 / 622 tok/s (the sparse indexer caps each row's keys past 4k) |
| 131,016-token chat question about its first page | answered exactly, 264 s end to end, at least 23% of memory free |
| decode without drafts, 20 / 600 / 4,200 / 32,768 tokens of context | 29.6 / 21.9 / 14.7 / 14.5 tok/s |
| MTP drafts against "draft": false (replies token-identical) | code 25.7 to 44.6 tok/s (70% accepted), prose 26.4 to 34.8 (56%), math 26.2 to 46.0 (75%) |
| an image, 1200 x 800 PNG (six 504-px crops and the overview, 692 prompt tokens) | 2.1 s to a 4-token answer |
| an image, 600 x 400 PNG (overview only, 193 prompt tokens) | 0.7 s |